

An
ABSTRACT
ON
Blockchain-Based Academic Credential Verification with
Issuer Accreditation Tracking

BACHELOR OF ENGINEERING
IN
COMPUTER SCIENCE AND ENGINEERING

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ABSTRACT

Educational and professional credential fraud remains a significant global concern, with fraudulent institutions and forged certificates undermining trust in academic qualifications. Traditional paper-based verification methods are slow and vulnerable to forgery, while existing blockchain-based verification systems, though effective at proving that a certificate has not been tampered with after issuance, do not adequately verify whether the issuing institution itself was accredited at the time of issuance. This gap allows unrecognized or fraudulent institutions to produce certificates that appear technically valid despite lacking legitimate accreditation. This project proposes a blockchain-based framework for academic credential verification that incorporates continuous issuer accreditation tracking alongside standard tamper-evidence guarantees. The system is designed with a hierarchical structure comprising regulatory authorities, accredited institutions, and certificate holders, implemented through smart contracts deployed on a public blockchain test network. Certificate files are stored off-chain using the InterPlanetary File System (IPFS), with only cryptographic hashes and accreditation metadata recorded on-chain to ensure both cost efficiency and tamper-evidence. A dedicated accreditation-tracking mechanism was developed to reflect changes in an institution's regulatory status, ensuring that verification results remain accurate even if accreditation is revoked after certificates were issued. Multi-signature issuance was additionally incorporated to prevent certificate forgery arising from a single compromised institutional credential. The system was implemented using Solidity for smart contract development, Hardhat for testing and deployment, IPFS for decentralized file storage, and React.js with Ethers.js for the frontend interface. The proposed framework enables fast, cost-free verification while addressing a critical limitation identified in existing credential verification systems, offering a more robust and India-relevant solution for combating credential fraud across a fragmented, multi-regulator educational landscape.

Keywords: Blockchain, Smart Contracts, Credential Verification, Issuer Accreditation, IPFS, Multi-Signature Authentication.